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Question # 01

Solution

First of all we calculate gradient of $\varphi = px^2y + qyz^2 + rzx^2y$

$$\varphi = px^2y + qyz^2 + rzx^2y$$

$$\nabla\varphi = \left(\frac{\partial\varphi}{\partial x}\hat{i} + \frac{\partial\varphi}{\partial y}\hat{j} + \frac{\partial\varphi}{\partial z}\hat{k}\right)$$

$$\frac{\partial\varphi}{\partial x} = \frac{\partial}{\partial x}(px^2y + qyz^2 + rzx^2y)\hat{i} = (2pxy + 2rzxy)\hat{i}$$

$$\frac{\partial\varphi}{\partial y} = \frac{\partial}{\partial y}(px^2y + qyz^2 + rzx^2y)\hat{j} = (px^2 + qz^2 + rzx^2)\hat{j}$$

$$\frac{\partial\varphi}{\partial z} = \frac{\partial}{\partial z}(px^2y + qyz^2 + rzx^2y)\hat{k} = (2qyz + rx^2y)\hat{k}$$

$$\nabla\varphi = \left((2pxy + 2rzxy)\hat{i} + (px^2 + qz^2 + rzx^2)\hat{j} + (2qyz + rx^2y)\hat{k}\right)$$

at the point (2,-1,3) the value of gradient is

=(-4p-12r)i+(4p+9q+12r)j+(-6q-4r)k

the maximum directional derivative parallel to x-axis if

as given magnitude of this maximum dirctional derivative is 72

so

-4p-12r = 72 . . . (1) along x-axis

4p+9q+12r = 0 . . . .(2) along y-axis

-6q-4r = 0 . . . (3) along z-axis

by solving eq (1),(2)and (3)

we get the values of p,q and r

p=-18

$q = 8$

$r = -12$

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Question # 02

Solution

$w = (2x - y)i + (x + bz)\,j + (y + 4z)\,k$

for solenoidal

$\nabla . w = 0$

$$\nabla . w = \left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k} \right) . \left((2x - y)i + (x + bz)\,j + (y + 4z)\,k \right)$$

$$= \frac{\partial}{\partial x}(2x - y)i + \frac{\partial}{\partial y}(x + bz)\,j + \frac{\partial}{\partial z}(y + 4z)k$$

$$= 2 + 0 + 4$$

$$= 6$$

$\sin ce$ the divergence is 6

so the vector field cannot be solenoidal for any choice of b.

the given verctor field is not solenoidal

